DAY-12 HDOC

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Batch- 15

1. Algorithm

Main Algorithm

1. Start.
2. Input a positive integer `number`.
3. If `number <= 0`, display an invalid input message and stop.
4. Display the menu:
   - **1:** Find the sum and product of the first and last digits.
   - **2:** Count digits lesser than, equal to, and greater than `k`.
5. Input the user's `choice`.
6. Use a `switch-case` statement to perform the selected operation.
7. Stop.

Case 1: Sum and Product of First and Last Digits

1. Find the last digit:
   `last = number % 10`
2. Store `number` in `temp`.
3. Repeatedly divide `temp` by 10 until `temp < 10`.
4. The remaining value of `temp` is the first digit.
5. Calculate:
   `sum = first + last`
6. Calculate:
   `product = first * last`
7. Display the sum and product.

Case 2: Compare Each Digit with `k`

1. Input digit `k`.
2. Check that `k` is between 0 and 9.
3. Initialize:
   `lesser = 0`, `equal = 0`, `greater = 0`
4. Store `number` in `temp`.
5. Repeat while `temp > 0`:
   - Find digit using `digit = temp % 10`.
   - If `digit < k`, increment `lesser`.
   - Else if `digit == k`, increment `equal`.
   - Else, increment `greater`.
   - Remove the last digit using `temp = temp / 10`.

6. Display all three counts.

2. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Number = 12345, Choice =	Sum = 6, Product = 5	Sum = 6, Product = 5	**Pass**
2	Number = 12345, Choice = 2, k = 3	Lesser = 2, Equal = 1, Greater = 2	Lesser = 2, Equal = 1, Greater = 2	**Pass**
3	Number = 50705, Choice =	Sum = 10, Product = 25	Sum = 10, Product = 25	**Pass**
4	Number = 55521, Choice = 2, k = 5	Lesser = 2, Equal = 3, Greater = 0	Lesser = 2, Equal = 3, Greater = 0	**Pass**

3. C Program

```
#include <stdio.h>

int main()

{

    long long number, temp;

    int choice;

    int first, last, digit, k;

    int lesser = 0, equal = 0, greater = 0;

   printf("Enter a positive integer: ");

    scanf("%lld", &number);

    if (number <= 0)

    {

       printf("Invalid input. Please enter a positive integer.\n");

        return 0;

    }

    printf("\n----- MENU -----\n");

    printf("1. Find sum and product of first and last digits\n");

    printf("2. Count digits lesser than, equal to, and greater than k\n");
```

```
printf("Enter your choice: ");

scanf("%d", &choice);

switch (choice)

{

    case 1:

        /* Find last digit */

        last = number % 10;

        /* Find first digit */

        temp = number;

        while (temp >= 10)

        {

            temp = temp / 10;

        }

          first = temp;

        printf("\nFirst digit = %d\n", first);

        printf("Last digit = %d\n", last);

        printf("Sum = %d\n", first + last);

        printf("Product = %d\n", first * last);

        break;

    case 2:

        printf("Enter digit k (0-9): ");

        scanf("%d", &k);

       if (k < 0 || k > 9)

        {

            printf("Invalid digit k.\n");

            break;
```

```c
        }
        temp = number;
        while (temp > 0)
        {
            digit = temp % 10;
            if (digit < k)
            {
                lesser++;
            }
            else if (digit == k)
            {
                equal++;
            }
            else
            {
                greater++;
            }
            temp = temp / 10;
        }
        printf("\nCount_lesser = %d\n", lesser);
        printf("Count_equal = %d\n", equal);
        printf("Count_greater = %d\n", greater);
        break;
    default:
        printf("Invalid choice.\n");
}
```

return 0;

}

```
UW PICO 5.09                          File: day12.c                          Modified

#include <stdio.h>

int main()
{
    long long number, temp;
    int choice;
    int first, last, digit, k;
    int lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
    scanf("%lld", &number);

    if (number <= 0)
    {
        printf("Invalid input. Please enter a positive integer.\n");
        return 0;
    }

    printf("\n----- MENU -----\n");
    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to, and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            /* Find last digit */
            last = number % 10;

            /* Find first digit */
            temp = number;

            while (temp >= 10)
            {
                temp = temp / 10;
            }

            first = temp;

            printf("\nFirst digit = %d\n", first);
            printf("Last digit = %d\n", last);
            printf("Sum = %d\n", first + last);
            printf("Product = %d\n", first * last);

            break;

        case 2:
            printf("Enter digit k (0-9): ");
            scanf("%d", &k);

            if (k < 0 || k > 9)
            {
                printf("Invalid digit k.\n");
                break;
            }

            temp = number;

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

```
UW PICO 5.09                          File: day12.c                          Modified
            /* Find first digit */
            temp = number;

            while (temp >= 10)
            {
                temp = temp / 10;
            }

            first = temp;

            printf("\nFirst digit = %d\n", first);
            printf("Last digit = %d\n", last);
            printf("Sum = %d\n", first + last);
            printf("Product = %d\n", first * last);

            break;

        case 2:
            printf("Enter digit k (0-9): ");
            scanf("%d", &k);

            if (k < 0 || k > 9)
            {
                printf("Invalid digit k.\n");
                break;
            }

            temp = number;

            while (temp > 0)
            {
                digit = temp % 10;

                if (digit < k)
                {
                    lesser++;
                }
                else if (digit == k)
                {
                    equal++;
                }
                else
                {
                    greater++;
                }

                temp = temp / 10;
            }

            printf("\nCount_lesser = %d\n", lesser);
            printf("Count_equal = %d\n", equal);
            printf("Count_greater = %d\n", greater);

            break;

        default:
            printf("Invalid choice.\n");
    }

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

```
  UW PICO 5.09                         File: day12.c                         Modified

        {
            digit = temp % 10;

            if (digit < k)
            {
                lesser++;
            }
            else if (digit == k)
            {
                equal++;
            }
            else
            {
                greater++;
            }

            temp = temp / 10;
        }

        printf("\nCount_lesser = %d\n", lesser);
        printf("Count_equal = %d\n", equal);
        printf("Count_greater = %d\n", greater);

        break;

    default:
        printf("Invalid choice.\n");
  }

  return 0;
}

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

4. Compilation and execution

All 2 Cases Executed successfully

```
Last login: Sat Sep  5 14:52:17 on ttys000
aaryanrajput@aaryans-macbook ~ % nano day11.c
aaryanrajput@aaryans-macbook ~ % clang day11.c -o day11.c
aaryanrajput@aaryans-macbook ~ % ./day11.c
Enter a positive integer: 12345

----- MENU -----
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1

First digit = 1
Last digit = 5
Sum = 6
Product = 5
aaryanrajput@aaryans-macbook ~ % ./day11.c
Enter a positive integer: 12345

----- MENU -----
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0-9): 3

Count_lesser = 2
Count_equal = 1
Count_greater = 2
aaryanrajput@aaryans-macbook ~ %
```

Final Result: The program compiled successfully and produced the expected results for **all 4 prepared test cases**.